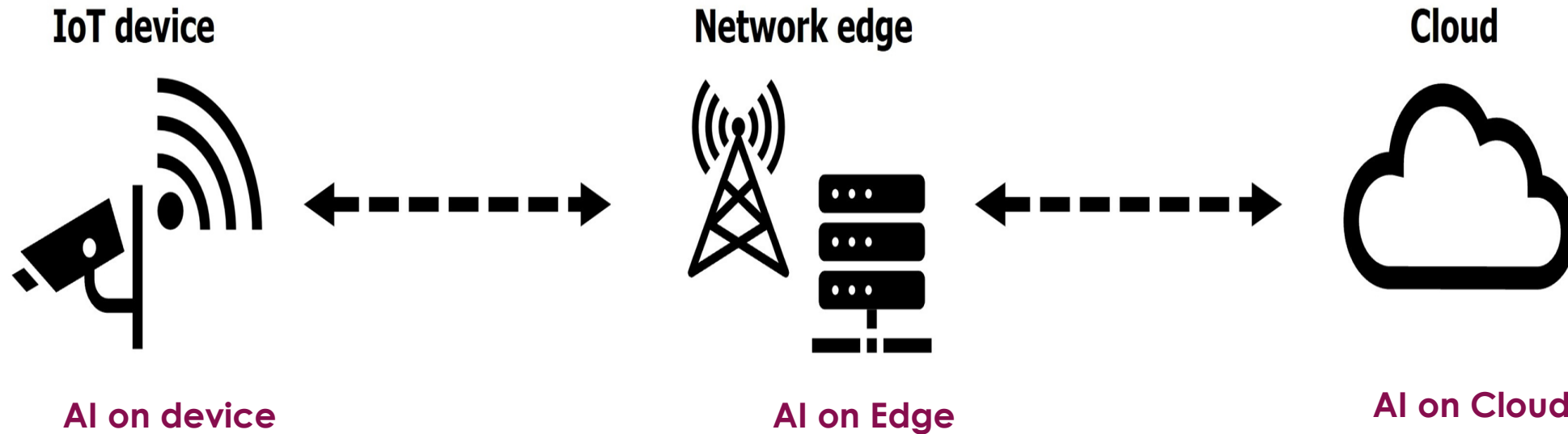


Cloud AIoT Reference Architecture

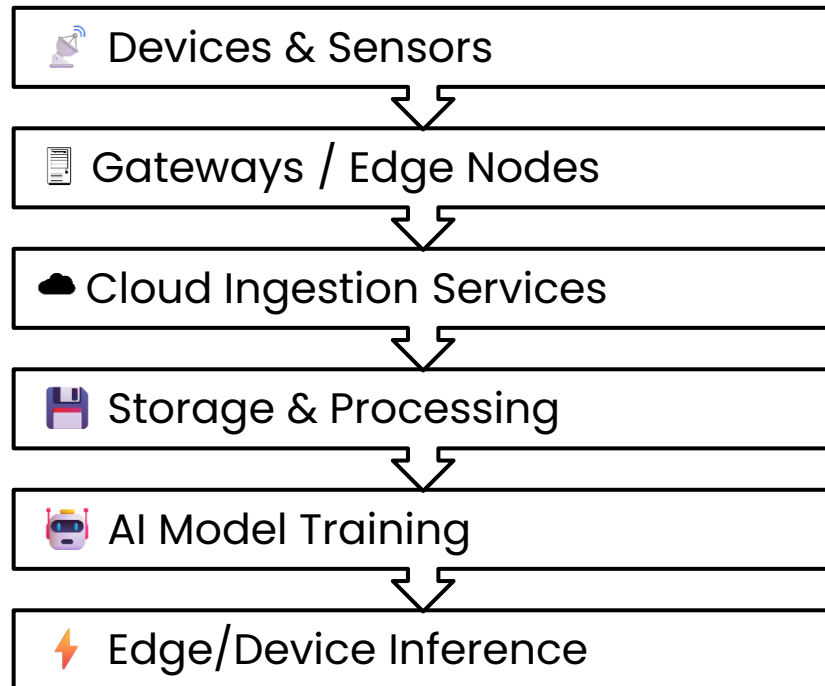


What is Cloud AIoT Architecture?

A unified platform connecting devices, edge systems, cloud services, AI, and applications.

Cloud AIoT Reference Architecture

Data Flow in Cloud AIoT



Role of the Cloud



Centralized intelligence & control



Scalable Storage & compute power



Advanced analytics & AI services

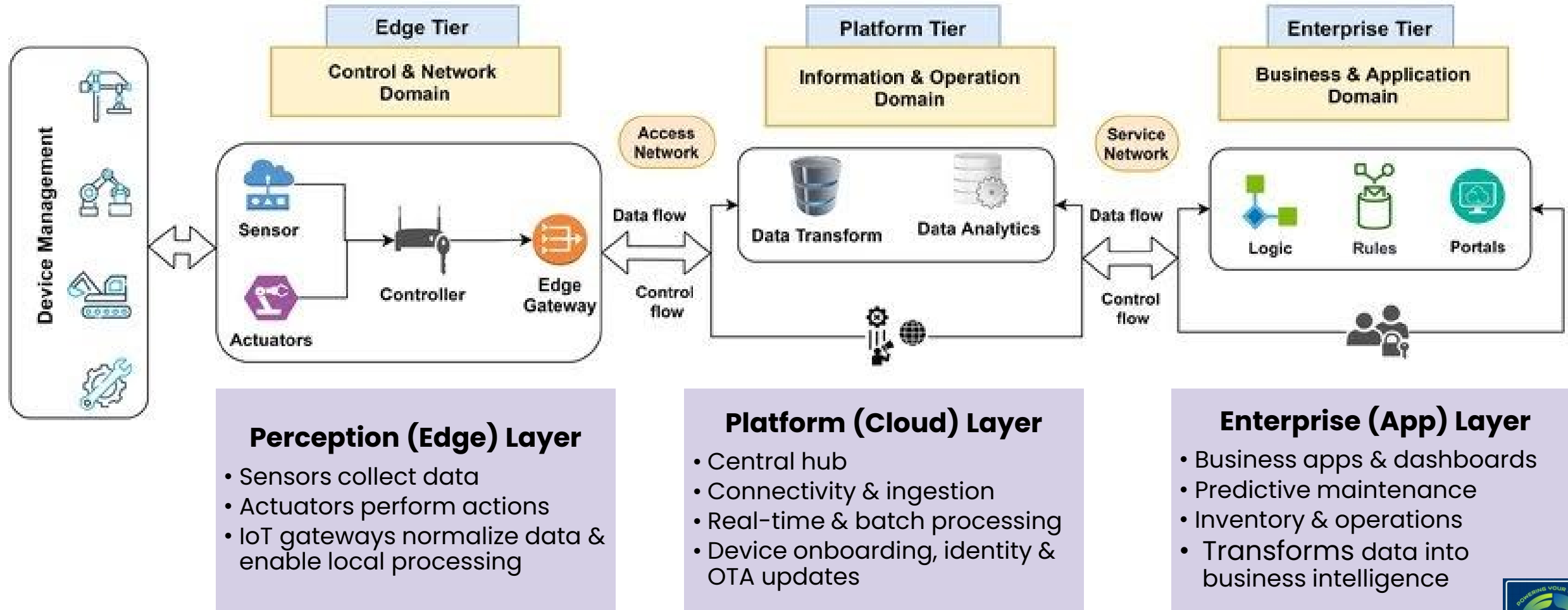


Device & model lifecycle management

Cloud acts as the intelligence hub for large-scale AIoT systems

Major Industry Reference Models

Most AIoT reference models follow a **three-tier architecture**: Edge → Platform → Enterprise



Core Architecture Layers in Cloud AIoT

Reference models accelerate design, interoperability, and scalability

Industrial & International Standards



Industrial Internet Reference Architecture (IIRA) – Designed for high-reliability industrial systems



ISO/IEC 30141 – Common vocabulary & reusable IoT design patterns

Vendor-Specific Architectures



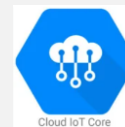
Microsoft Azure IoT

Azure IoT Hub for device connectivity
Digital Twins for system modelling & control



AWS IoT

AWS IoT Core for secure device communication
Event-driven analytics & automation



Google Cloud IoT Architectures

Pub/Sub-based data ingestion
Scalable analytics & data pipelines

Key Design Principles for AIoT Architecture

Security by Design

- Apply security at every layer
- Device identity & authentication
- Encrypted communication
- Access control & trust management

Scalability

- Support growth in devices & data volume
- Modular architecture & automation
- Elastic cloud resources & orchestration

Interoperability

- Use standardized protocols & open standards
- Enable multi-vendor device compatibility
- Ensure seamless system integration

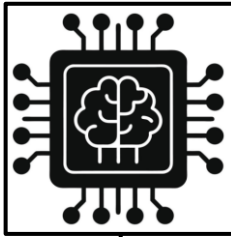
Emerging Architectural Trends

Modern architectures enable real-time intelligence and operational visibility



Digital Twins

- Virtual replicas of physical assets
- Simulate behavior & predict failures
- Enable predictive maintenance & optimization



Intelligent Edge

- Process data closer to the source
- Reduce latency & bandwidth usage
- Enable real-time control & mission-critical decisions



Unified Namespace (UNS)

- Central data broker with standardized naming
- Creates a single source of truth
- Enables seamless enterprise-wide data access